

CREX Polarimetry

2022 Hall A Winter Collaboration Meeting

Eric King

On behalf of the CREX Møller and Compton Polarimetry Working Groups



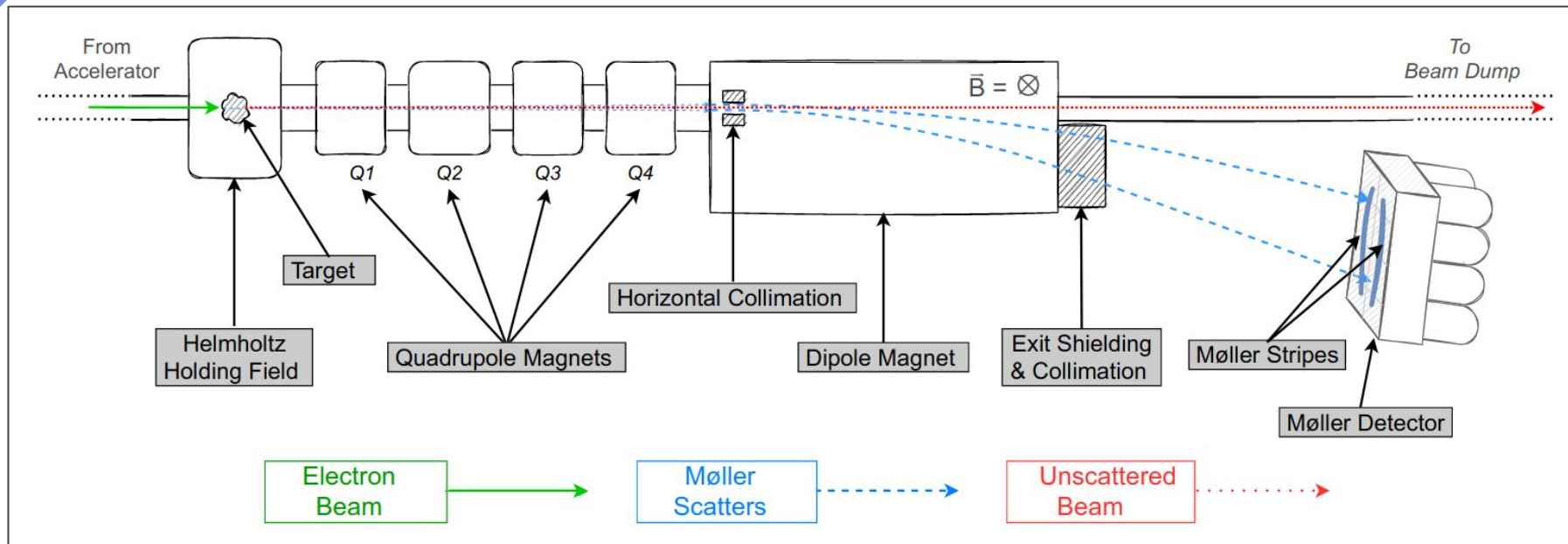
TEMPLE
UNIVERSITY

 **Jefferson Lab**



Polarimeters

Hall A Møller Polarimeter



- Beam scatters off polarized foil.
- Mollers are steered by quadrupoles into dipole.
- Dipole bends events to detector situated below the beamline.

Hall A Møller Polarimeter

Polarized
Cross section

Unpolarized
Cross-section

Analyzing
Powers

Utilizes Møller scattering
(elastic ee)

Measures large $\sim 5\%$ QED
asymmetry for polarized scatters

$$\frac{d\sigma^M}{d\Omega} = \frac{d\sigma_0^M}{d\Omega} \left(1 + \sum P_{beam,i} P_{targ,j} A_{ij} \right)$$

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Brute Force Setup

$P_{targ} \parallel P_{beam}$



$$A_{zz} = - \frac{\sin^2 \theta^* (7 + \cos^2 \theta^*)}{(3 + \cos^2 \theta^*)^2} \approx - \frac{7}{9} \quad \text{Around } 90^\circ \text{ COM scatters}$$

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Polarized
Cross section

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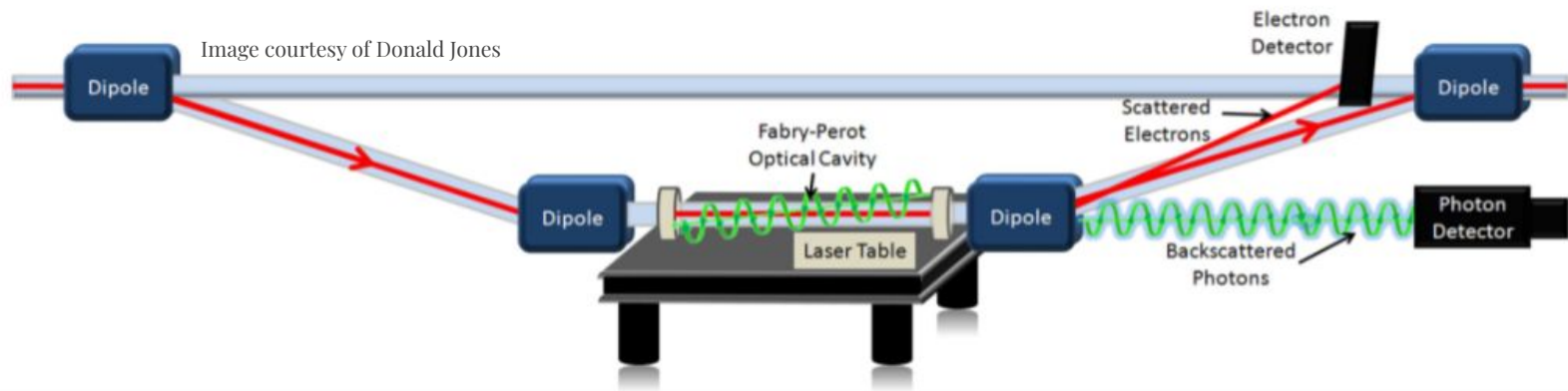
Measured
asymmetry

Target polarization
known

Computed
analyzing
power

$$\frac{\frac{d\sigma}{d\Omega}_a - \frac{d\sigma}{d\Omega}_b}{\frac{d\sigma}{d\Omega}_a + \frac{d\sigma}{d\Omega}_b} = A_{meas} = P_{beam} P_{targ} \langle A_{zz} \rangle$$

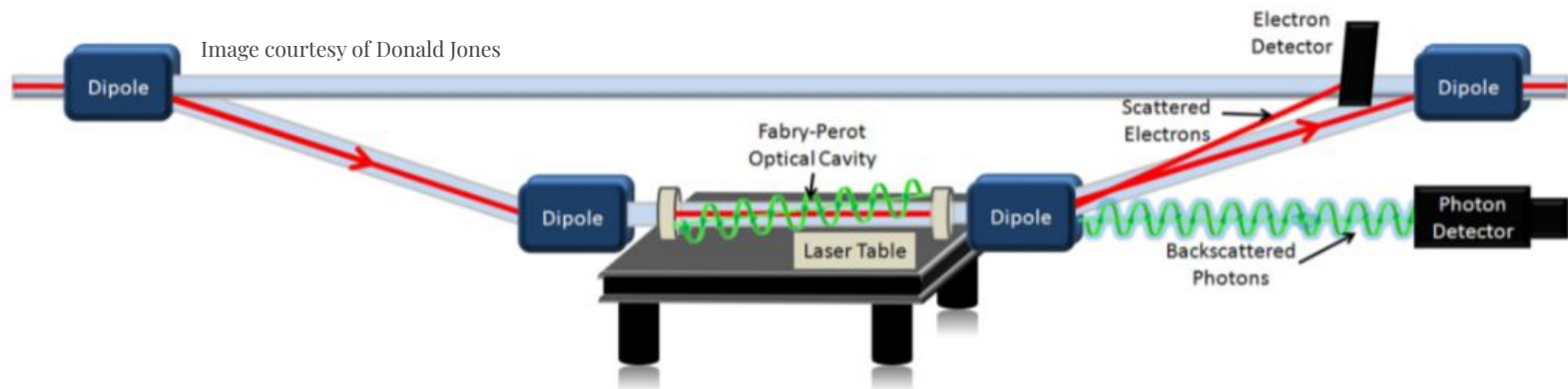
Compton Polarimeter



Takes advantage of the scattering asymmetry of Compton Scattering

- Beam electrons interact with γ in optical cavity.
- Back-scattered photons picked up by photon detector.
- Compton polarimetry allows a non-invasive continuous measurement of beam polarization while experiments are running.

Compton Polarimeter



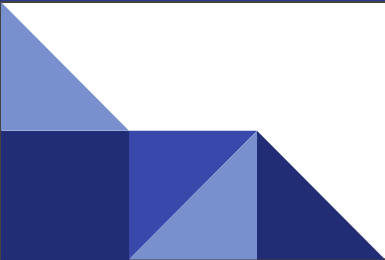
In order to subtract out backgrounds, laser alternates between on and off phases.

Measured asymmetry

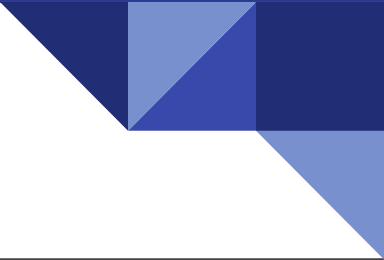
γ are $\sim$ circularly polarized

Computed longitudinal analyzing power

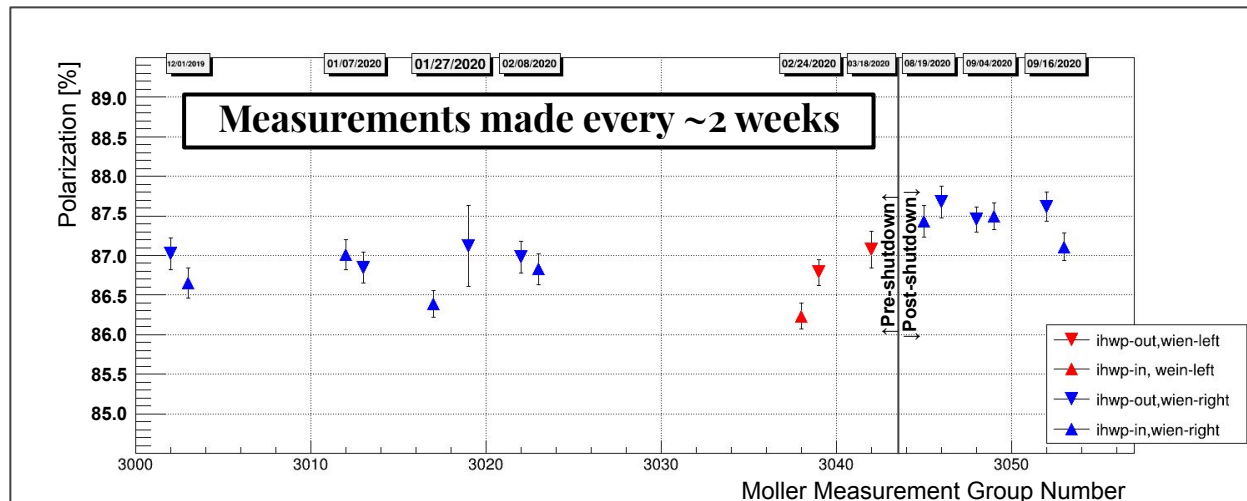
$$\langle \mathcal{A}_{ON} \rangle - \langle \mathcal{A}_{OFF} \rangle = \mathcal{A}_{exp} = P_{beam} P_{\gamma} \mathcal{A}_{\mathcal{L}}$$



Polarimetry Results



Møller Measurements During CREX



Data set:	All		Pre-Shutdown		Post-Shutdown	
Møller Mean Pol	87.08%	± 0.06%	86.72%	± 0.08%	87.43%	± 0.08%

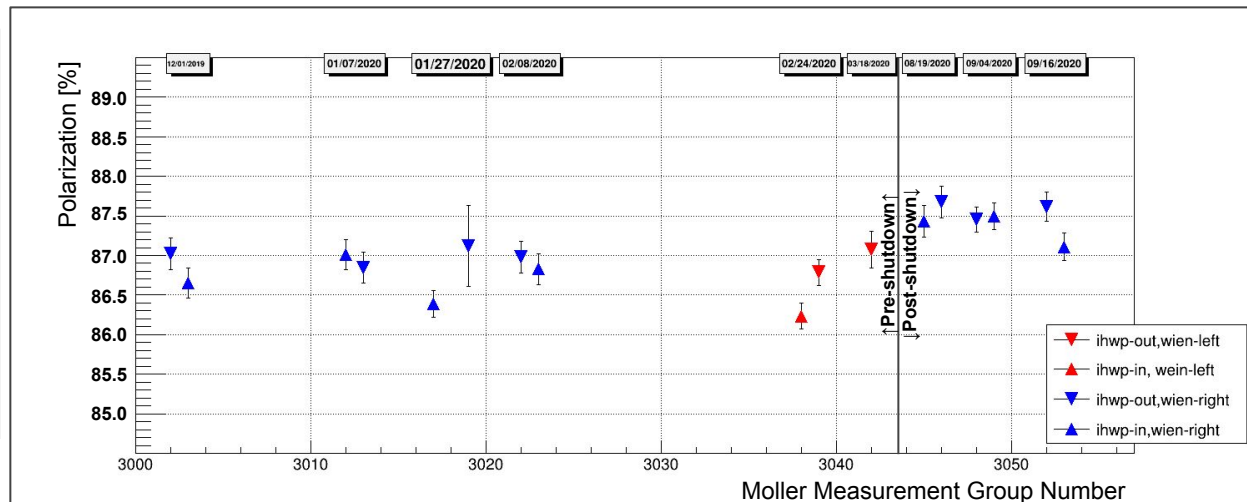
Møller Measurements During CREX

Systematics Table

Source	Value	$\delta P/P$ (%)
A_{zz}	0.75421	0.16
Foil Polarization	0.08005	0.57
Dead Time Correction	0.148%	0.15
Accidental Correction	0.205%	0.04
Charge Normalization	0.029%	0.01
Null Asymmetry (Cu Foil)	0.0%	0.22
PITA Variation	—	0.06
Spin Precession (dP/P)	—	0.04
High Current Extrapolation	—	0.50
Bleed through	—	0.18
Slit Dependence	—	0.10
Total		0.85

Total Møller systematic uncertainty during CREX: **0.85%**

Majority contributors are foil polarization uncertainty and high-current extrapolation uncertainty.



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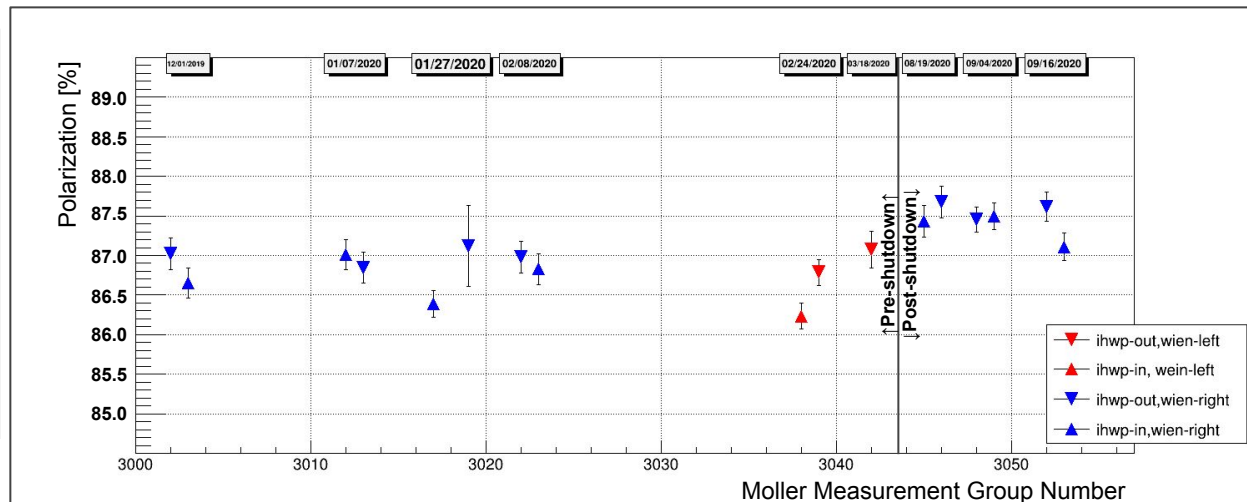
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Møller Mean Pol	87.08% $\pm$ 0.06%	86.72% $\pm$ 0.08%	87.43% $\pm$ 0.08%

➤ **Møller mean polarization over CREX was 87.08% $\pm$ 0.06% (stat) $\pm$ 0.85% (sys)**

CREX Systematics Comparison to PREX-I

Systematics Table CREX

Source	Value	$\delta P/P$ (%)
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Systematics Table PREX-I

Variable	dA / A
Foil Polarization	0.25%
Target Saturation	0.3%
Target Temperature	0.02%
Target-to-target variation	0.5%
Analyzing Power	0.3%
Levchuk Effect	0.5%
Dead time	0.3%
Background	0.3%
Others	0.5%
Total	1.1%

0.39%

Data: O. Glamazdin,
PREX Collaboration Meeting (2011)

- Quoted CREX “Foil Polarization” systematic: polarization, saturation and temperature corrections. We took a very conservative approach to this but it remains a dominant systematic.

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Data: O. Glamazdin,
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- [Major Improvement] Levchuk effect: Rolled into our analyzing power systematic.
 - We had an effective method for effectively eliminating this and the total systematic uncertainty for the Levchuk Effect was **0.06%**

CREX Systematics Comparison to PREX-I

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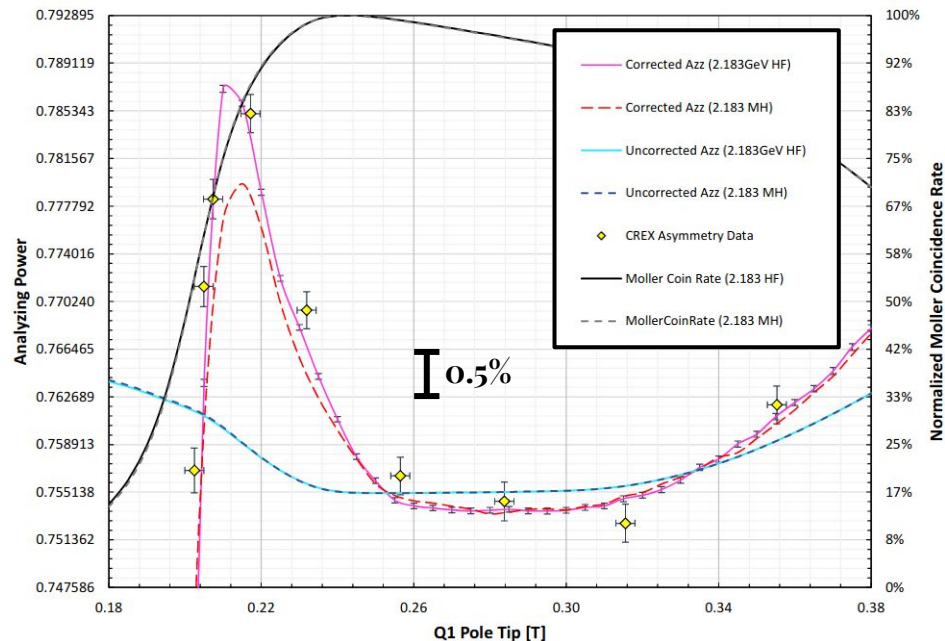
Data: O. Glamazdin,
PREX Collaboration Meeting (2011)

- High Current Extrapolation: Experiment runs at ~100 uA and Moller polarimetry is performed at ~1 uA range.
 - Constrained by 2007 Hall C studies to the 0.5% level.
 - Unsure if this was considered in PREX-I systematics.

Moller Improvements

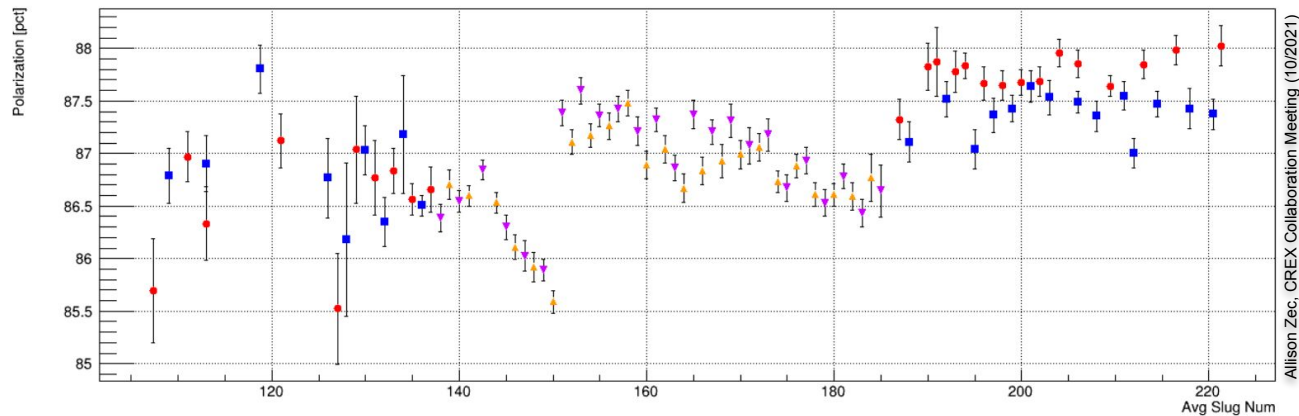
- Addition of harp for precision alignment of beam onto foil.
 - This made CREX Møller setups (and measurements) highly reproducible
- Granted systematics study time (image on right) that provided fundamental improvements on understanding our dominant analyzing power correction (Levchuk Effect).

CREX Q1 Scan :: 2.183GeV HartreeFock / 2.183GeV ModHy
[Anpower Bars are 0.5% Change off 0.755138]



- Modified Hydrogen wavefunctions ----- used for Levchuk Effect are now replaced by Hartree-Fock derived momentum distributions —————.

Compton Results During CREX



Allison Zec, CREX Collaboration Meeting (10/2021)

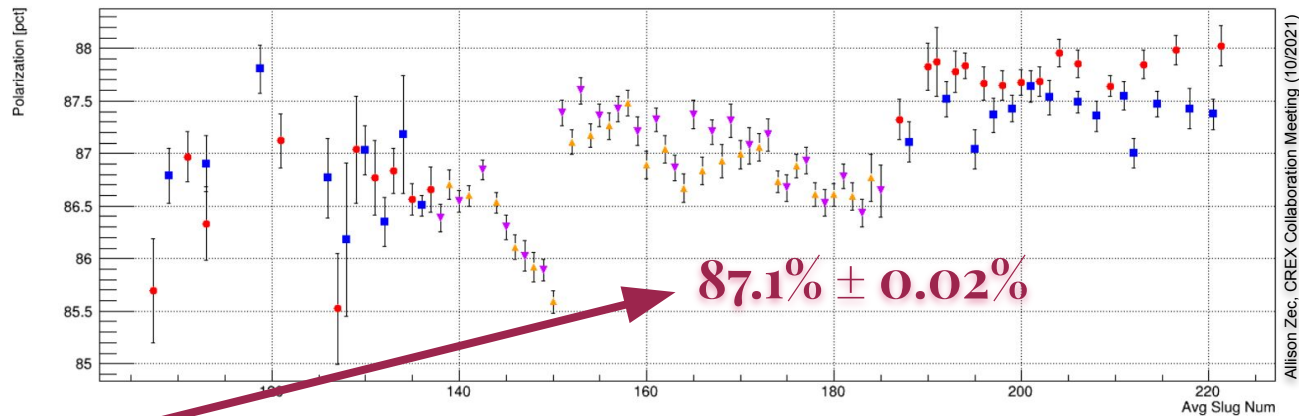
Compton Results During CREX

The Compton team examined multiple averaging models all of which yielded consistent results:

Escargatoire Average
 $87.118\% \pm 0.018\%$

Piecewise Fitting
 $87.119\% \pm 0.016\%$

Mini-Esc. Average
 $87.104\% \pm 0.019\%$

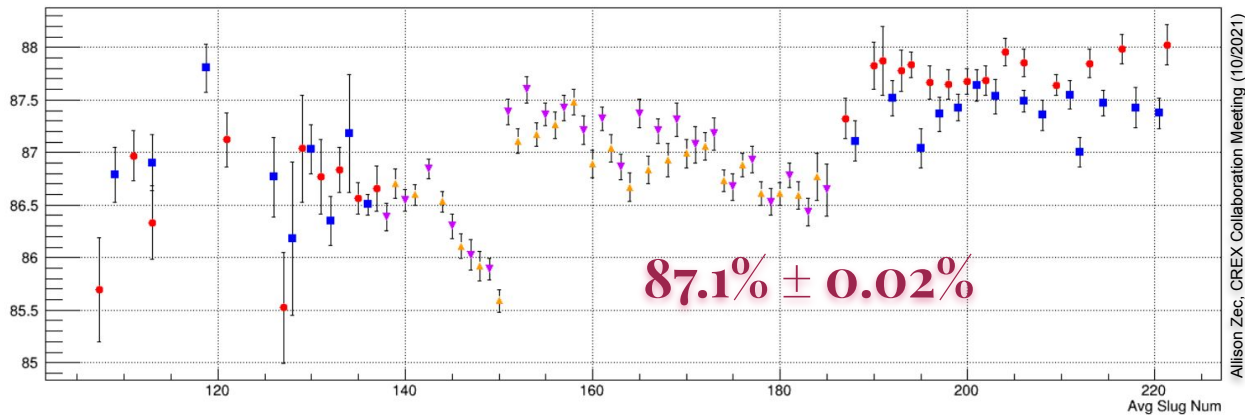


Compton Results During CREX

Compton Systematics Table

Source	Relative Correction	Uncertainty
Collimator	-	0.20%
Laser DOCP	0.29%	0.45%
Gain Shift	-	0.15%
Model	-	0.02%
Beam Energy	0.103%	0.05%
Nonlinearity	-	0.02%
Rad Corr	0.3%	< 0.01%
Statistics	-	0.02%
Total	-	0.52%

Data: Allison Zec, CREX Collaboration Meeting (10/2021)



Total Compton systematic was 0.52%

Driving contributor to Compton systematic is the degree of circular polarization at 0.45%.

■ Compton measurements aligned in time with CREX slugs

■ Compton polarization average weighted by A_{pV} uncertainty of matching slugs.

➤ Compton mean polarization over CREX was $87.1\% \pm 0.02\%$ (stat) $\pm 0.52\%$ (sys)

Compton Systematics Compared to HAPPEX-III

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Allison Zec, CREX Collaboration Meeting (10/2021)

HAPPEX-III Systematics Table

Systematic Errors	
Laser Polarization	0.80%
Signal Analyzing Power:	
Nonlinearity	0.30%
Energy Uncertainty	0.10%
Collimator Position	0.05%
Analyzing Power Total	0.33%
Gain Shift:	
Background Uncertainty	0.31%
Pedestal Uncertainty	0.20%
Gain Shift Total	0.37%
Total Uncertainty	0.94%

Data: M. Friend, et al, NIM A676 (2012) 96-105

- [Major Improvement] Measurement of the DOCP of the Compton laser.

Compton Systematics Compared to HAPPEX-III

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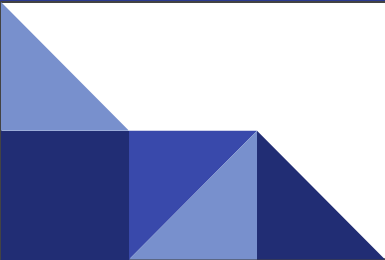
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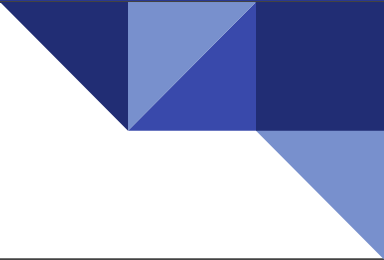
Data: M. Friend, et al, NIM A676 (2012) 96-105

- [Major Improvement #2] Better understanding of the photon detector gain shift.



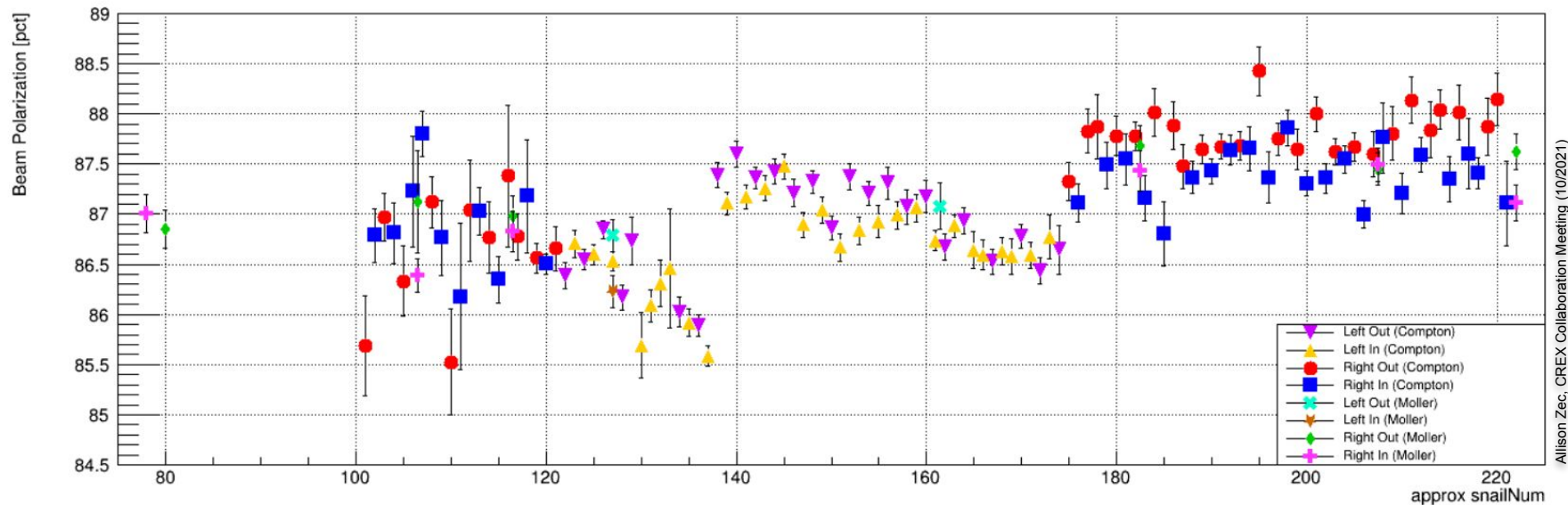
Final CREX Polarization

Moller/Compton Combined Results

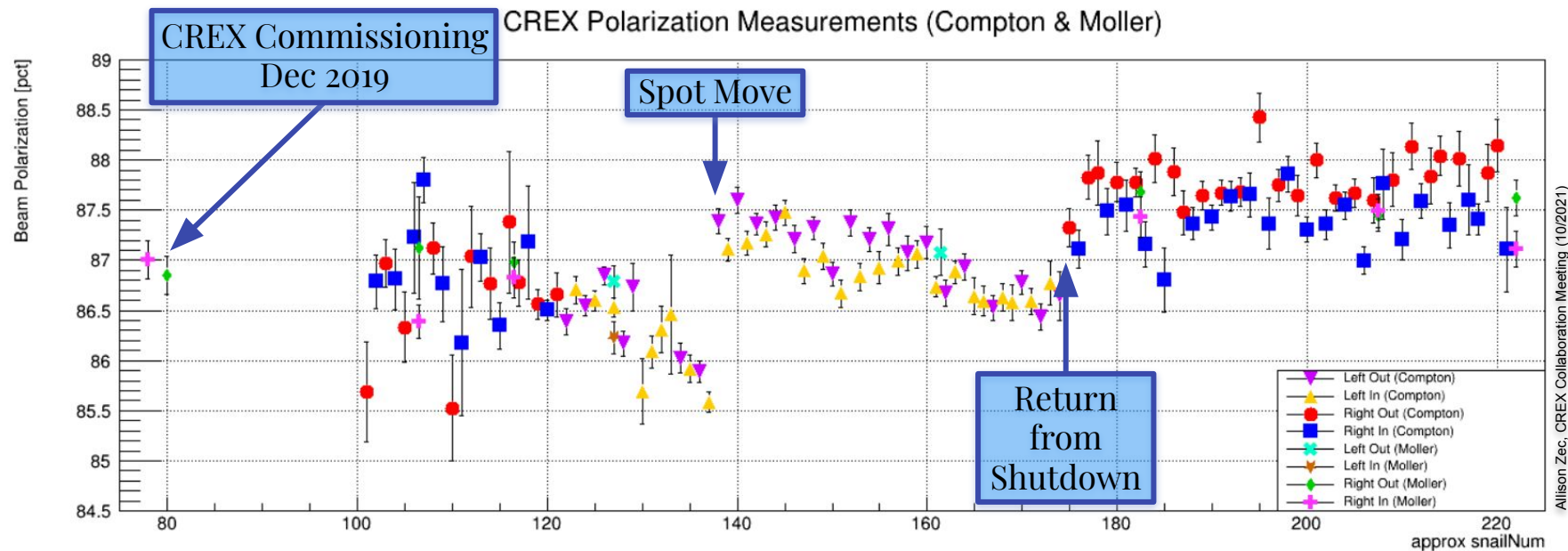


Compton and Møller Measurement Overlay

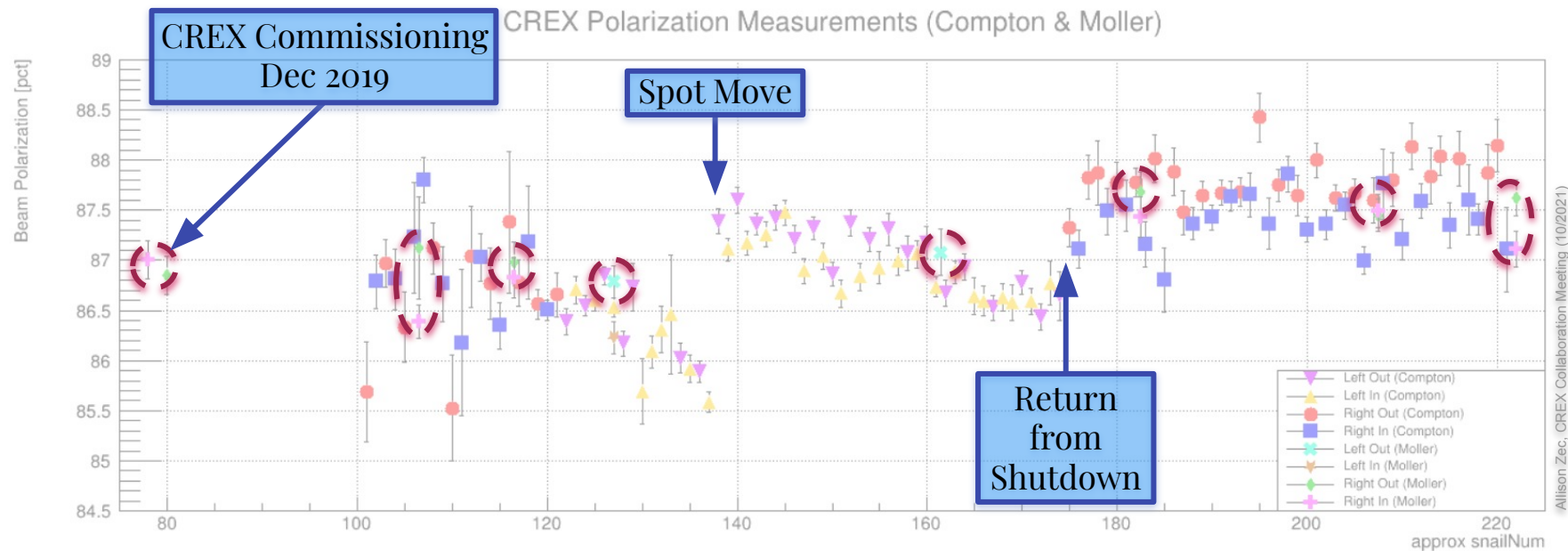
CREX Polarization Measurements (Compton & Moller)



Compton and Møller Measurement Overlay

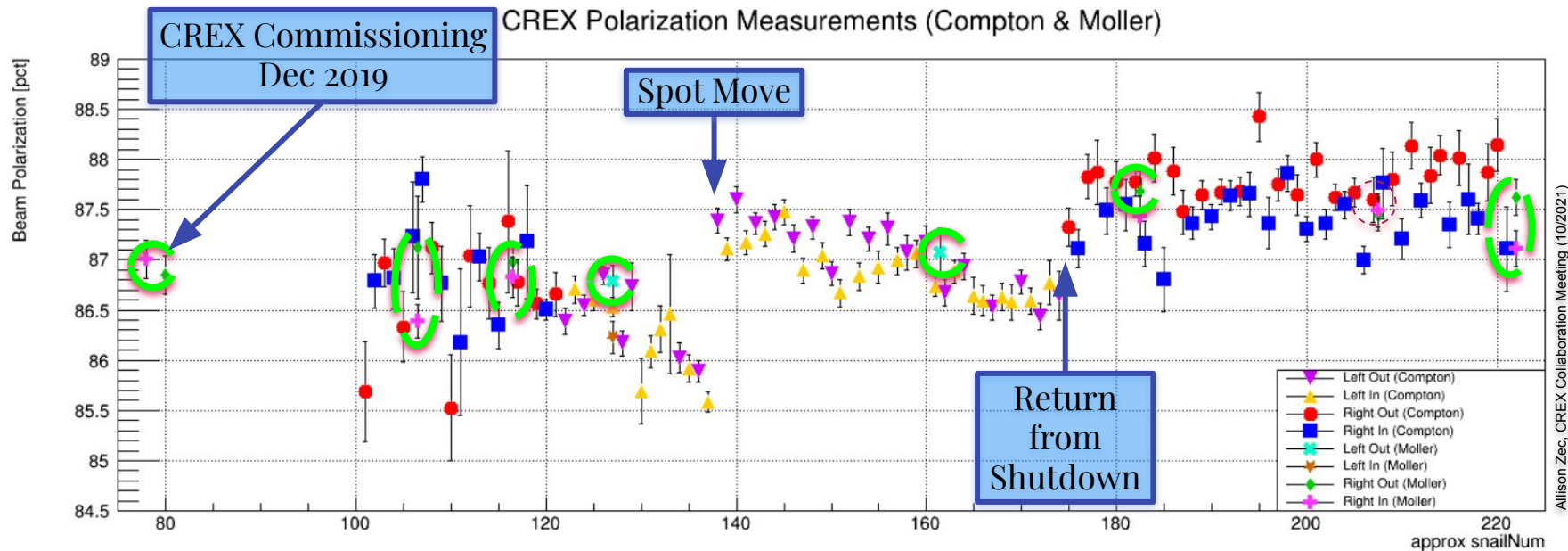


Compton and Møller Measurement Overlay



I've made an attempt to highlight the less-frequent Moller measurements among the Comptons

Compton and Møller Measurement Overlay

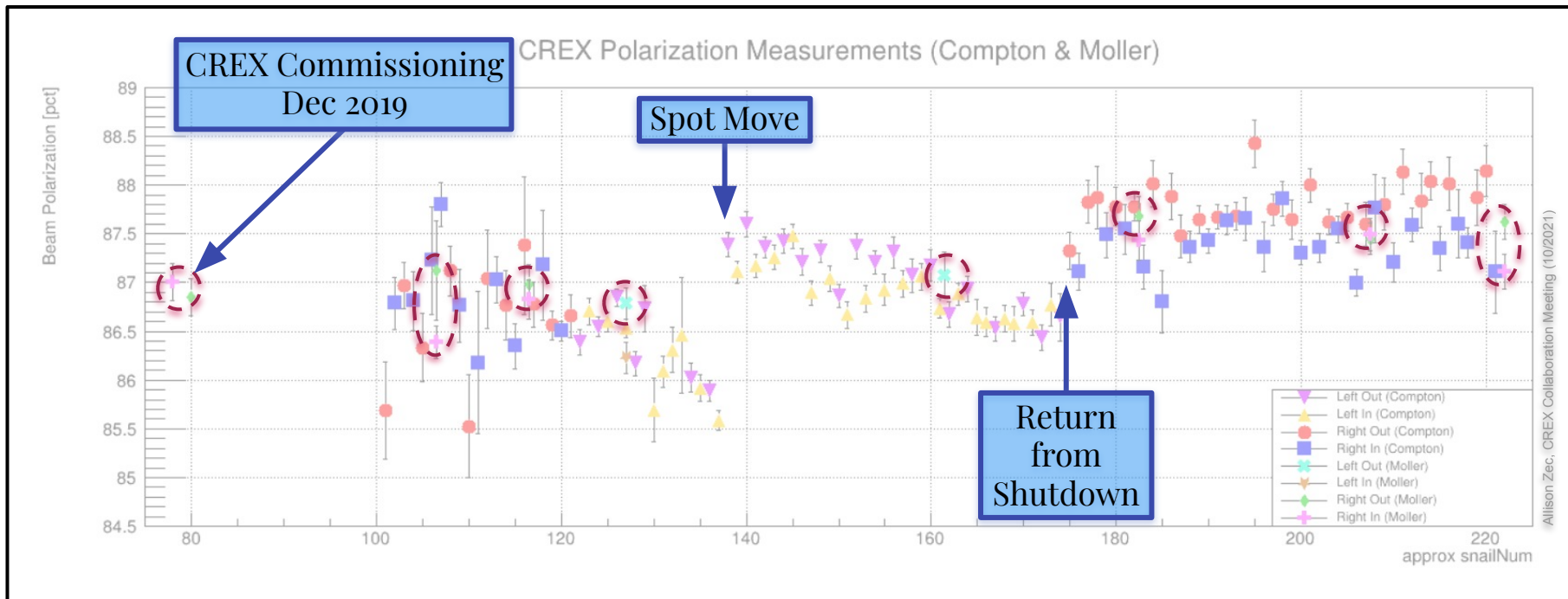


- Møller and Compton measurements were consistent throughout the CREX experiment.



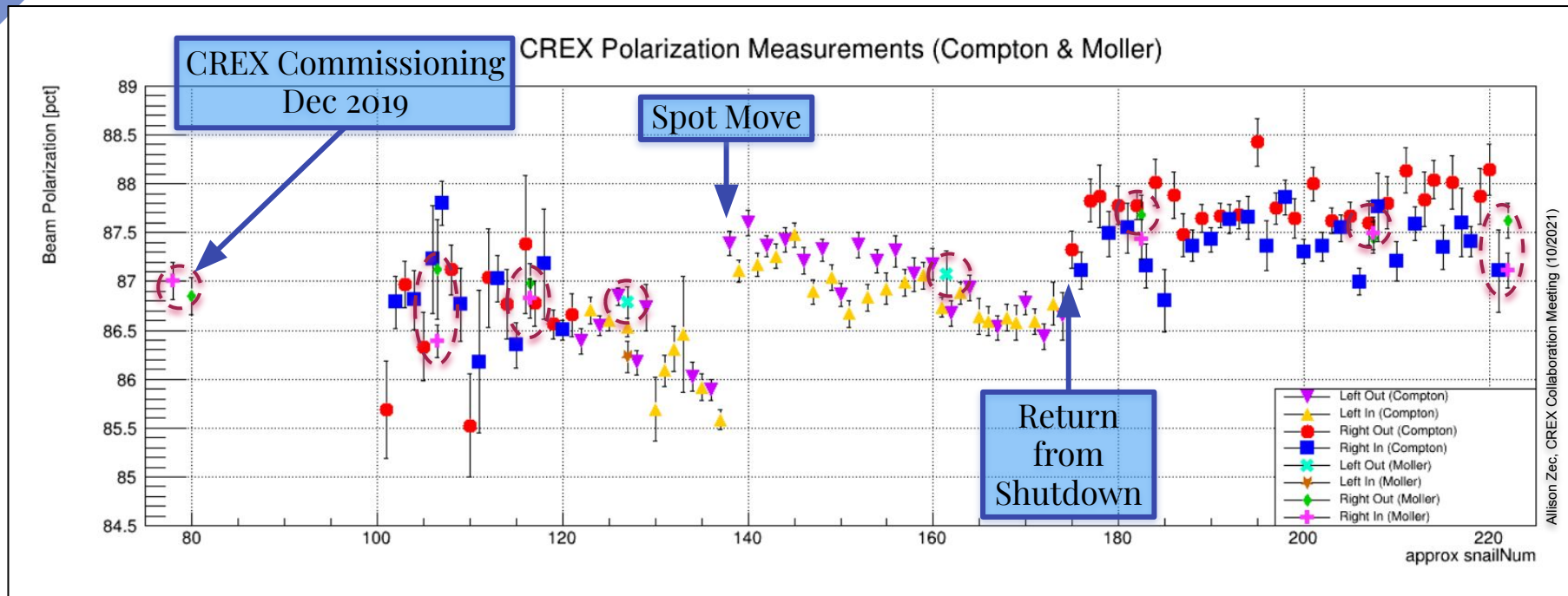
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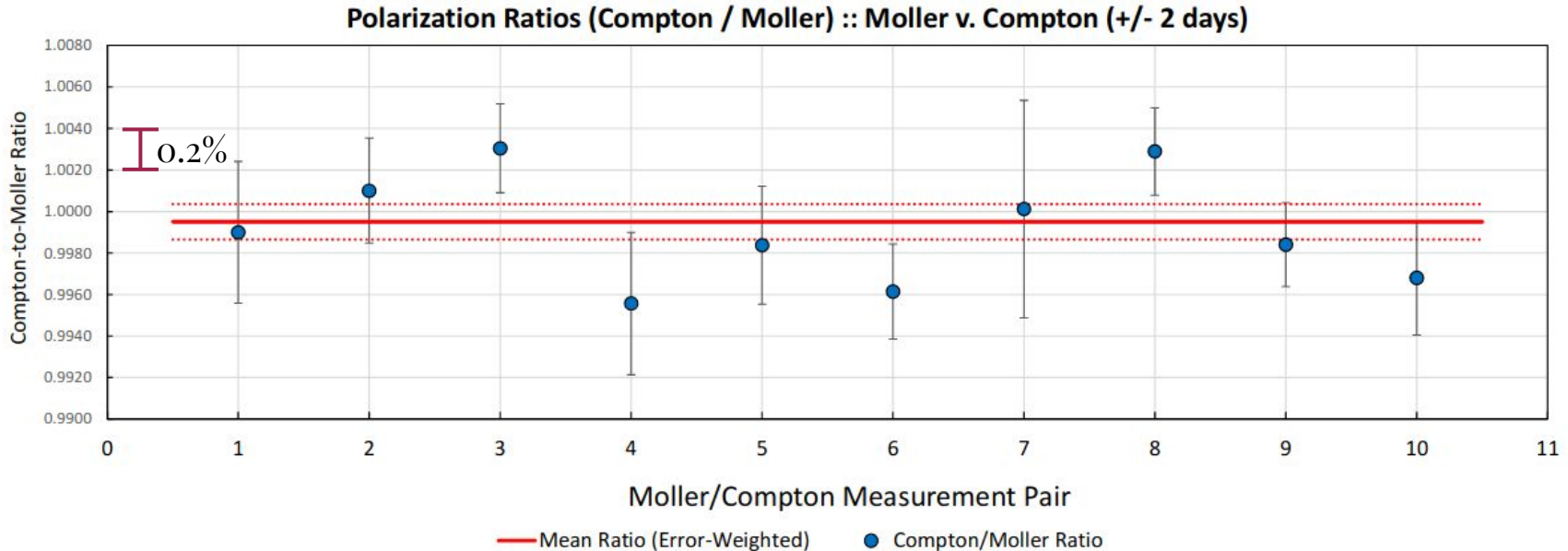
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Compton and Møller Measurement Overlay



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Comparison of Compton & Møller Polarization Measurements



➤ Møller measurements were compared to Compton measurements taken within roughly ± 48 hours.

➤ The mean Compton/Møller ratio was 0.9995 ± 0.0008
Ratio consistent with 1 at the $\sim 0.1\%$ level.

Final Combined Result for CREX

CREX Running Moller Result

$$\mathcal{P}_e = (87.1 \pm 0.85)\%$$

+

CREX Running Compton Result

$$\mathcal{P}_e = (87.1 \pm 0.52)\%$$

- Sub 1% precision measurement from Moller.
- This is a JLab tie for 'best' for a Møller measurement (between Hall A and C Møller polarimeters).
 - This is a marked improvement for the Hall A polarimeter.

- Highest precision for any experiment performed at JLab.
- Compton measurement at 0.5% is an *apparent* record breaker.
 - There's no knowledge of any experiment claiming better precision.

Final Combined Result for CREX

CREX Running Moller Result

$$\mathcal{P}_e = (87.1 \pm 0.85)\%$$

+

CREX Running Compton Result

$$\mathcal{P}_e = (87.1 \pm 0.52)\%$$

CREX Compton+Moller
Combined Result

$$\mathcal{P}_e = (87.1 \pm 0.44)\%$$

We have a combined
0.44% high-precision
polarimetry measurement!

- Substantial systematic uncertainty improvements
 - Compton: In high-precision territory 0.52%.
 - Møller: Improvements over previous measurements; current systematics are likely overly-conservative.
 - Major step towards future PV experimental requirements.
- Two independent measurements utilizing different physical processes



**Combined
0.44%**

THE TWO MEASUREMENTS AGREE !!!

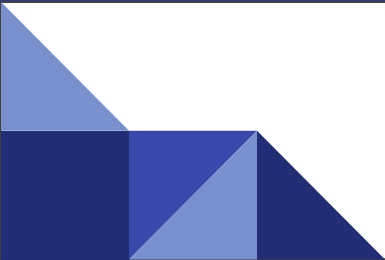
CREX Polarimetry Teams

Møller:

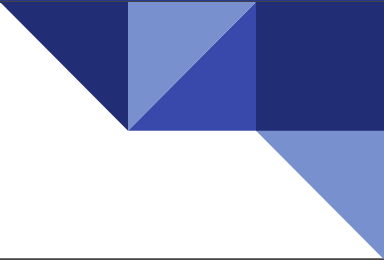
Eric King; Paul Souder; Donald Jones; Bill Henry; Jim Napolitano; Simona Malace; Dave Gaskell; and Kent Paschke.

Compton:

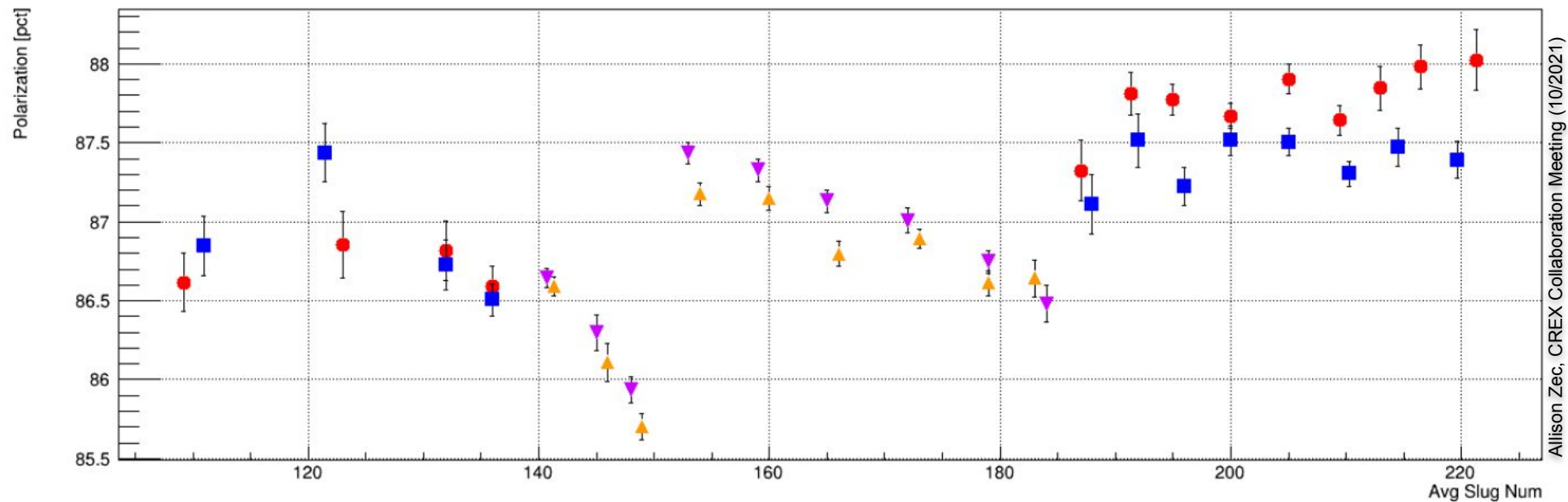
Allison Zec; Dave Gaskell; Amali Premithilake; Juan Carlos Cornejo; Kent Paschke; Ciprian Gal; Caryn Palatchi; and Mark Dalton.



Backup Slides



Compton Polarization Averaging [Method 1]

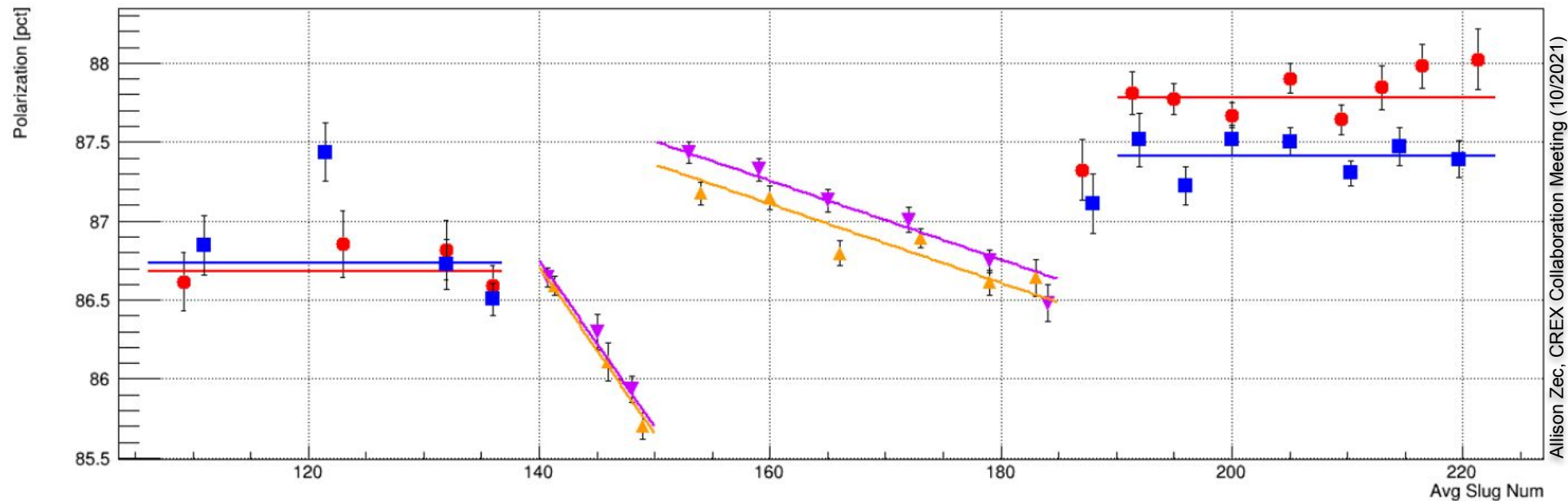


➤ Grouped Compton measurements which are aligned in time with CREX slugs.

➤ Groupings then weighted by error of corresponding A_{PV} measurement.

➤ Average calculated polarization:
 $(87.118 \pm 0.018 \text{ (stat)})\%$

Compton Polarization Averaging [Method 2]

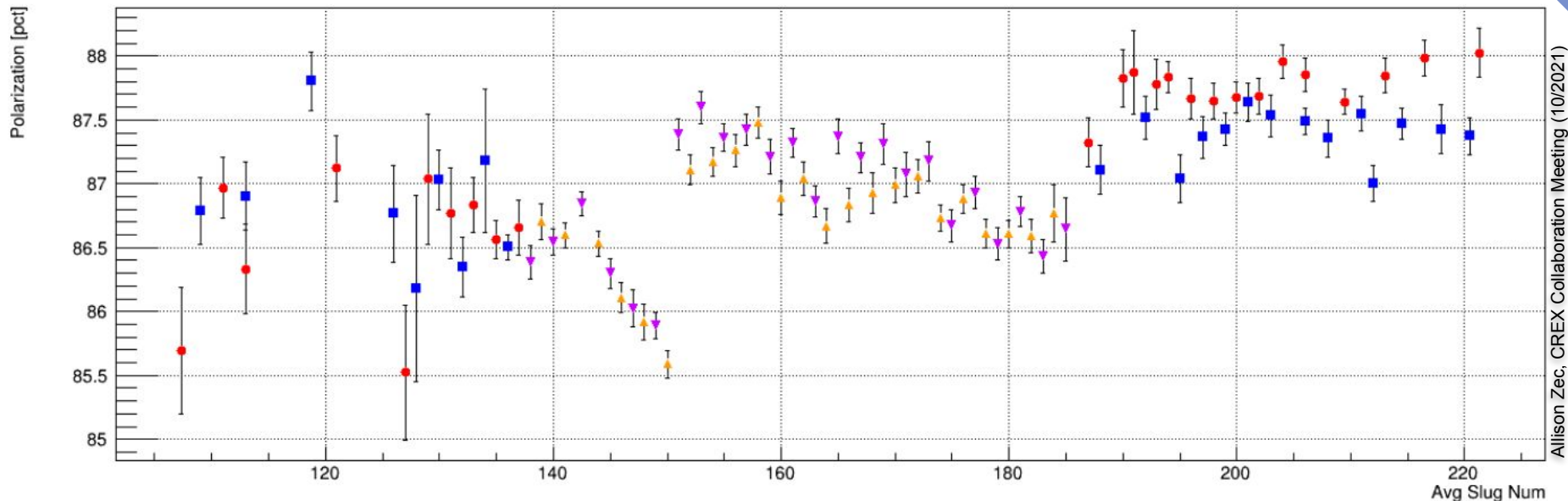


➤ Grouped Compton measurements which are aligned in time with CREX slugs.

➤ Groupings then weighted by error of corresponding A_{PV} measurement.

➤ Average calculated polarization:
 $(87.119 \pm 0.016 \text{ (stat)})\%$

Compton Polarization Averaging [Method 3]



- Compton measurements grouped IFF they overlapped with CREX slugs.
- Averages derived from fit evaluations and uncertainties come from fit parameters.
- Average calculated polarization:
 $(87.104 \pm 0.019 \text{ (stat)})\%$